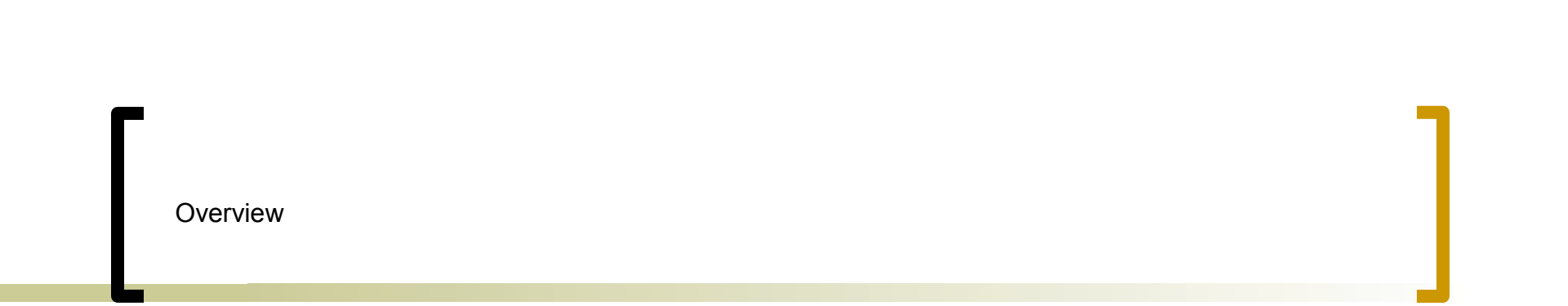




## FUNDAMENTAL OF INTERNET CONCEPTS

Learning objectives:-

1. Draw the block diagram of computer system showing various units
2. Outline various application of IT
3. Differentiate between assembly and high level language
4. List the features of the MS Word, MS-Excel & MS-PowerPoint
5. Demonstrate the use of PowerPoint for seminar presentation
6. Identify various web browser
7. Use internet and create mail ID, send and receive Email



## Overview

1. History of the Internet
2. Web Environment Overview
3. TCP/IP
4. HTTP
5. URL
6. DNS

## Definition of the Internet

■ An electronic network of computers that includes nearly every university, government, and research facility in the world. Also included are many commercial sites. It started with four interconnected computers in 1969 and was known as ARPAnet.

[www.orafaq.com/glossary/fagglosi.htm](http://www.orafaq.com/glossary/fagglosi.htm)

■ A network of computer networks which operates world-wide using a common set of communications protocols.

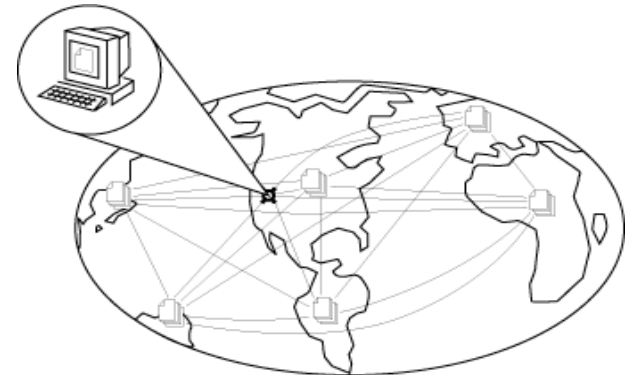
[www.utas.edu.au/library/etutor/main/webzglos.htm](http://www.utas.edu.au/library/etutor/main/webzglos.htm)

■ The vast collection of inter-connected networks across the world that all use the TCP/IP protocols.

[www.liv.ac.uk/webteam/glossary/](http://www.liv.ac.uk/webteam/glossary/)

## What Is the Internet?

- ✓ A network of networks, joining many government, university and private computers together and providing an infrastructure for the use of E-mail, bulletin boards, file archives, hypertext documents, databases and other computational resources
- ✓ The vast collection of computer networks which form and act as a single huge network for transport of data and messages across distances which can be anywhere from the same office to anywhere in the world.





## History of the Internet

# Internet Timeline

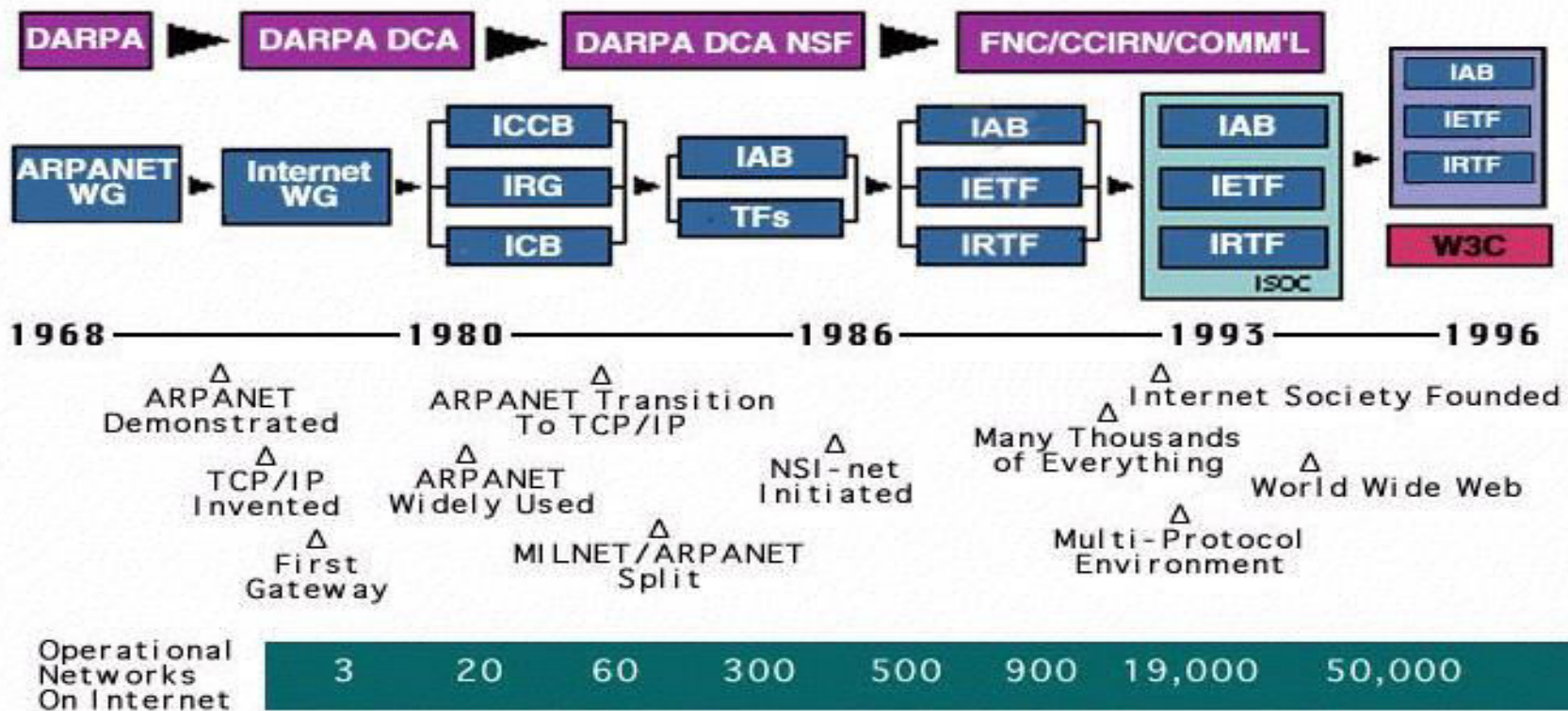
- 1958** President Dwight D. Eisenhower saw need for the Advanced Research Projects Agency (ARPA) to keep the U.S. at the forefront of technology
- 1962** Leonard Kleinrock invents packet-switching technology
- 1963** J.C.R. Licklider, head of computer research at ARPA, articulates vision of worldwide network
- 1967** Larry Roberts publishes a paper proposing the ARPAnet network
- 1968** DOD initiates the ARPAnet development
- 1969** ARPAnet unveiled at UCLA
- 1972** E-mail introduced by Ray Tomlinson
- 1988** Albert Gore, then a Tennessee senator, proposes the National Research and Education Network, which would provide top computing facilities to research communities and schools
- 1991** Gopher document retrieval system introduced at University of Minnesota
- 1992** The World Wide Web is born, introduced by Tim Berners-Lee. The first audio and video multicasts are broadcast over the Internet
- 1993** The Internet browser MOSAIC is introduced at the University of Illinois by Marc Andreessen
- 1994** Real Audio introduced to Internet which allows one to hear audio in near real time. Radio HK, first 24-hour Internet-only radio station, starts broadcasting
- 1996** Telecommunications Act of 1996 deregulates data network transmission
- 1999** 150 million users on the Internet. Over 800 million web pages accessible

## Brief History of the Internet

- ✓ 1968 - DARPA (Defense Advanced Research Projects Agency) contracts with BBN (Bolt, Beranek & Newman) to create ARPAnet
- ✓ 1970 - First five nodes:
  - UCLA
  - Stanford
  - UC Santa Barbara
  - U of Utah, and
  - BBN
- ✓ 1974 - TCP specification by Vint Cerf
- ✓ 1984 – On January 1, the Internet with its 1000 nodes converts en masse to using TCP/IP for its messaging



## Internet History



From Simple, But Significant Ideas Bigger Ones Grow 1940s to 1969

We will prove that packet switching  
works over a WAN.

Hypertext can be used to allow  
rapid access to text data

Packet switching can be used to  
send digitized data though  
computer networks

We can accomplish a lot by having a  
vast network of computers to use for  
accessing information and exchanging ideas

We can do it cheaply by using  
Digital circuits etched in silicon.

We do it reliably with “bits”,  
sending and receiving data

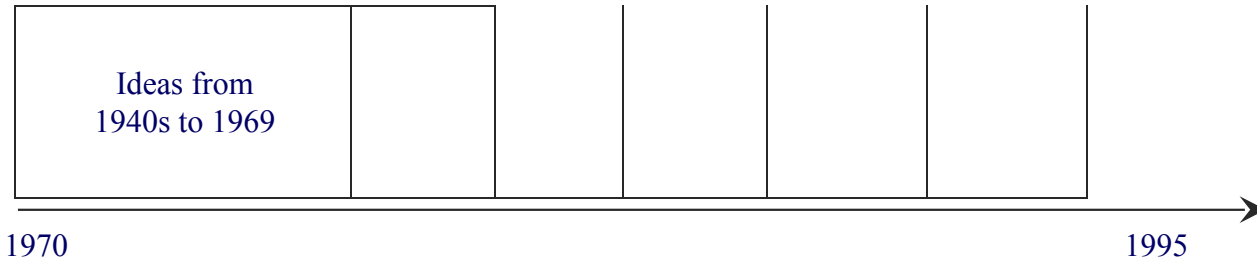
We can access  
information using  
electronic computers

1945

1969

## From Simple, But Significant Ideas Bigger Ones Grow 1970s to 1995

|                                                                                                                                                   |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
| Great efficiencies can be accomplished if we use<br>The Internet and the World Wide Web to conduct business.                                      |  |  |  |
| The World Wide Web is easier to use if we have a browser that<br>To browser web pages, running in a graphical user interface context.             |  |  |  |
| Computers connected via the Internet can be used<br>more easily if hypertext links are enabled using HTML<br>and URLs: it's called World Wide Web |  |  |  |
| The ARPANET needs to convert to<br>a standard protocol and be renamed to<br>The Internet                                                          |  |  |  |
| We need a protocol for Efficient<br>and Reliable transmission of<br>Packets over a WAN: TCP/IP                                                    |  |  |  |



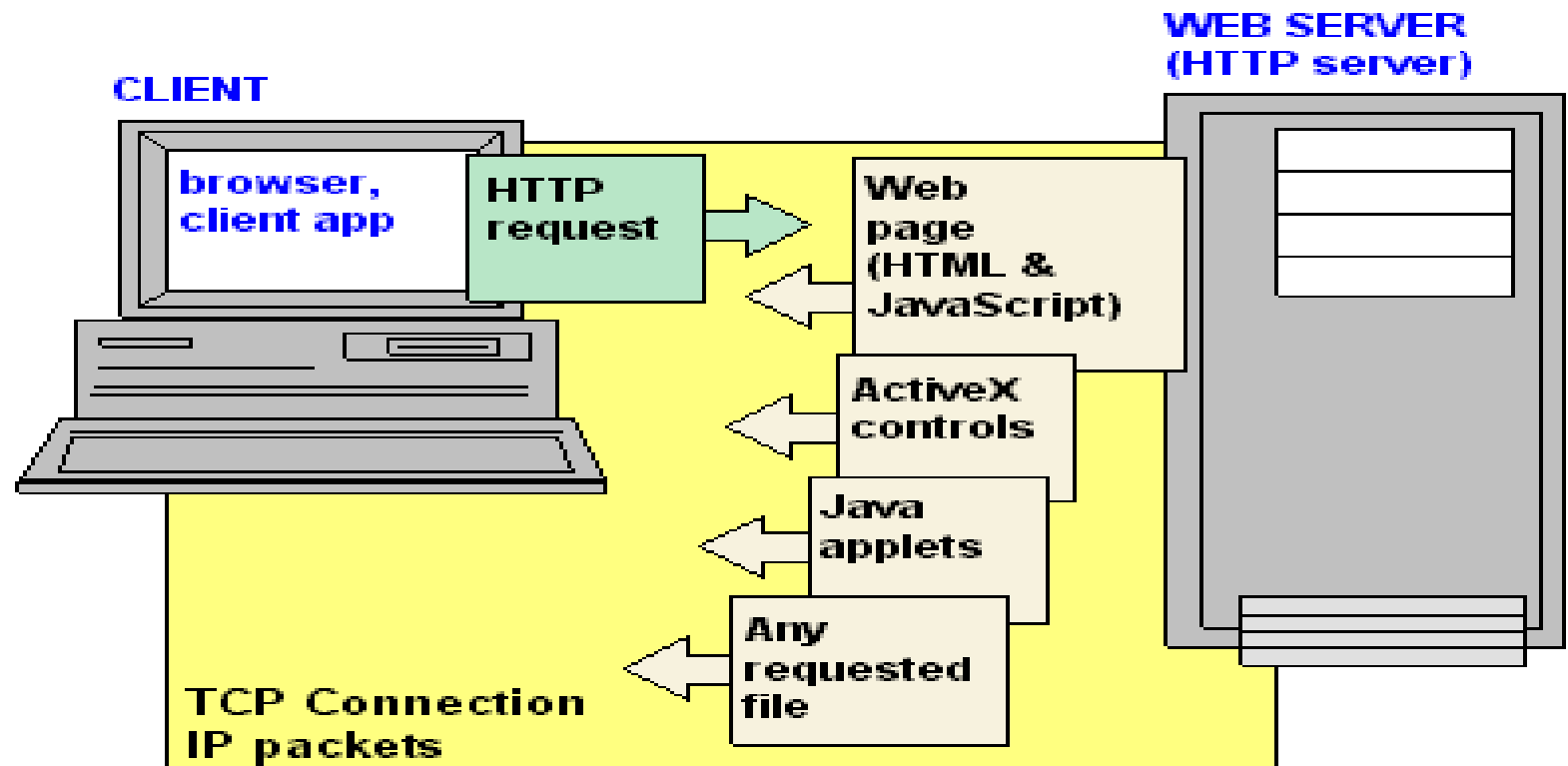
## The Creation of the Internet

- ✓ The creation of the Internet solved the following challenges:
- Basically inventing digital networking
  - Survivability of an infrastructure to send / receive high-speed electronic messages
  - Reliability of computer messaging



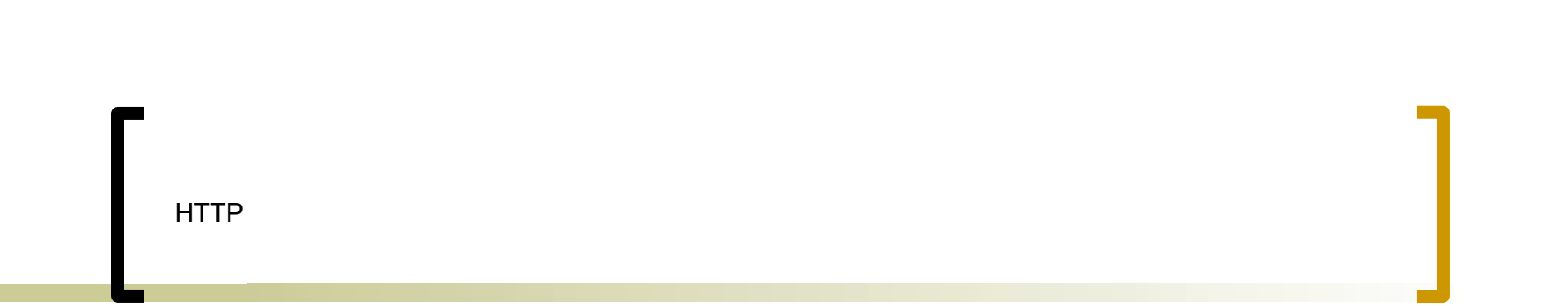
## Web Environment Overview

From Computer Desktop Encyclopedia  
© 2003 The Computer Language Co. Inc.



[ ... continued ]

- Web browsers communicate with Web servers via the TCP/IP protocol. The browser sends HTTP requests to the server, which responds with HTML pages and possibly additional programs in the form of ActiveX controls or Java applets.



HTTP

- A protocol used to request and transmit files, especially webpages and webpage components, over the Internet or other computer network.



URL

- Uniform Resource Locator, or URL, is a fancy name for the address of a web page on the world wide web.

## URL

- The first part of the URL is called the protocol. It tells the browser how to deal with the file that it is about to open. One of the most common protocols you will see is HTTP, or Hypertext Transfer Protocol.
- The second part of the URL is the name of the server where the file is located, followed by the path that leads to the file and the file's name itself, as illustrated below:

**"http://www.site.com/fun/file.html"**

| Protocol | Server Name  | Path  | File Name |
|----------|--------------|-------|-----------|
| http     | www.site.com | /fun/ | file.html |

- Sometimes, a URL ends in a trailing forward slash with no filename given, as below.
- **"http://www.site.com/fun/"**
- In this case the URL refers to the default file in the last directory in the path, which is a file named "index.html." An equivalent URL to the one above would be
- **"http://www.site.com/fun/index.html"**

## Absolute URLs

- A URL that specifies the full path to a document page, which includes the domain name and protocol.
- Also known as an "explicit URL".
- Example:
- <http://www.unikl.edu.my/network.htm>

## Relative URLs

- Relative URLs are analogous to giving directions to someone such as "go down the hall and turn right." In other words, the directions refer to where you are starting from. In the same way, a relative URL describes the location of the desired file with reference to the location of the file that contains the URL itself.
- For example, a relative URL for a file that is in the same directory as the current file (that is, the one with the link that points to that file) is merely the file name and extension, such as **"index.html"**
- You create a URL for a file in a subdirectory of the current directory by placing the name of the subdirectory first and following it with a forward slash and then the name and extension of the desired file, as shown below.

Inside the current folder...

**"fun/file.html"**

...there is a folder called "fun"...

...that contains a file named "file.html"

## Relative URLs

- To reference a file in a directory at a higher level of the file hierarchy, use two periods ( .. ) as shown below.



You can combine and repeat the two periods and forward slash to reference any file on the same server as the current file.

## DNS: Domain Name System

- A system for converting host names and domain names into IP addresses on the Internet or on local networks that use the TCP/IP protocol. For example, when a Web site address is given to the DNS either by typing a URL in a browser or behind the scenes from one application to another, DNS servers return the IP address of the server associated with that name.
- In this hypothetical example, [www.company.com](http://www.company.com) would be converted into the IP address 204.0.8.51. Without DNS, you would have to type the four numbers and dots into your browser to retrieve the Web site, which of course, you can do.

## DNS Hierarchy

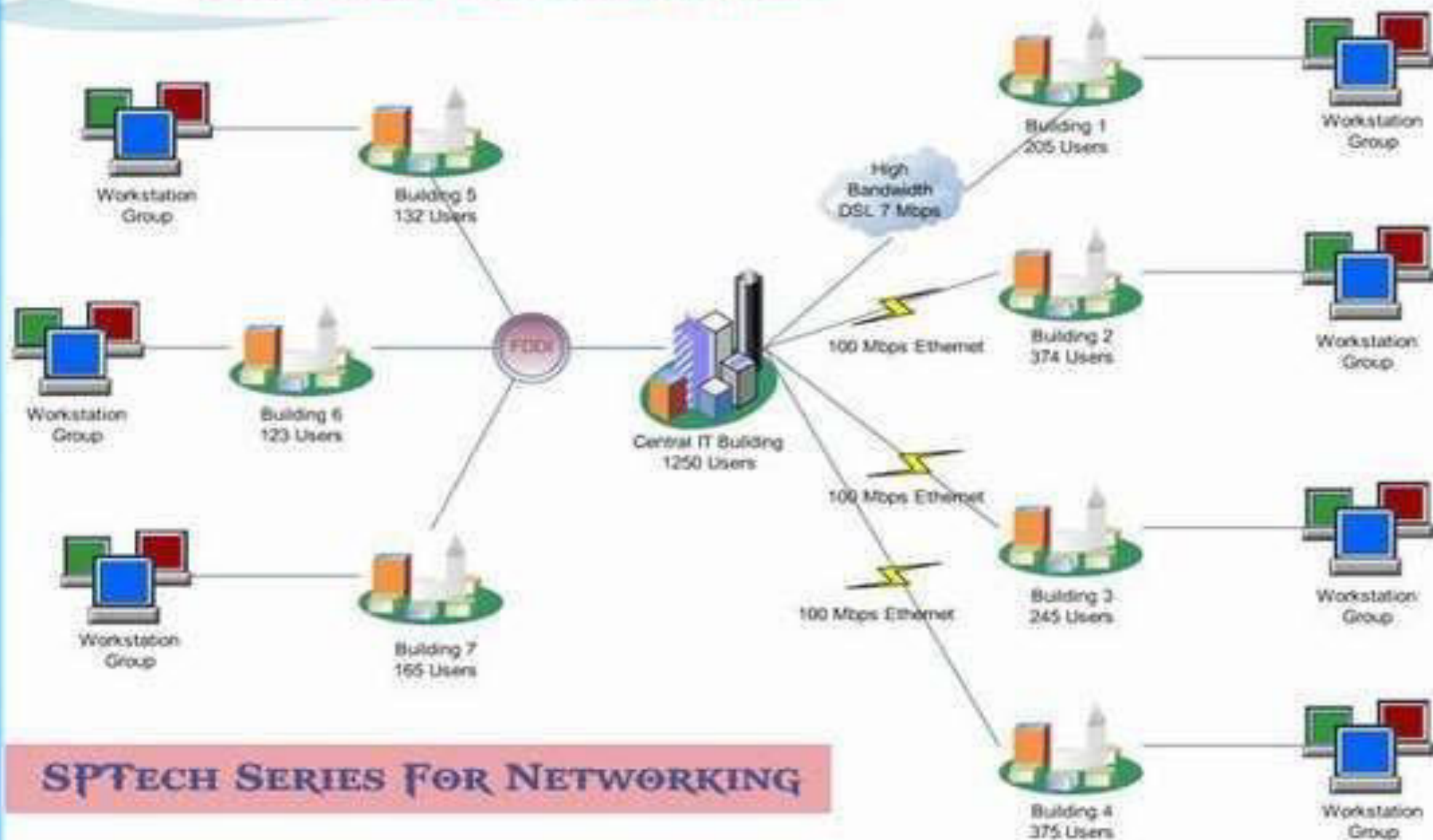
- The DNS system is a hierarchy of database servers that start with the root servers for all the top level domains (.com, .net, etc.).
- The root servers point to authoritative servers residing within ISPs and companies that resolve the host names to complete the name resolution.
- Using the example [WWW.COMPANY.COM](http://WWW.COMPANY.COM), COMPANY.COM is the domain name, and WWW is the host name. The domain name is the organization's identity on the Web, and the host name is the name of the actual Web server within that domain



## INTRODUCTION

- A computer network is a set of computers connected together for the purpose of sharing resources. The most common resource shared today is connection to the Internet. Other shared resources can include a printer or a **file server**. The Internet itself can be considered a computer network.
- Computer networks support an enormous number of applications and services such as access to the World Wide Web, digital video, digital audio, shared use of application and storage servers, printers, and fax machines, and use of email and instant messaging applications as well as many others. Computer networks differ in the transmission medium used to carry their signals, communications protocols to organize network traffic, the network's size, topology, traffic control mechanism and organizational intent. The best-known computer network is the Internet.

## COMPUTER NETWORK TYPES



## LOCAL AREA NETWORK (LAN)

- A *local-area network (LAN)* is a computer network that spans a relatively small area. Most often, a LAN is confined to a single room, building or group of buildings, however, one LAN can be connected to other LANs over any distance via telephone lines and radiowaves.
- A system of LANs connected in this way is called a wide-area network (WAN). The difference between a LAN and WAN is that the wide-area network spans a relatively large geographical area. Typically, a WAN consists of two or more local-area networks (LANs) and are often connected through public networks.

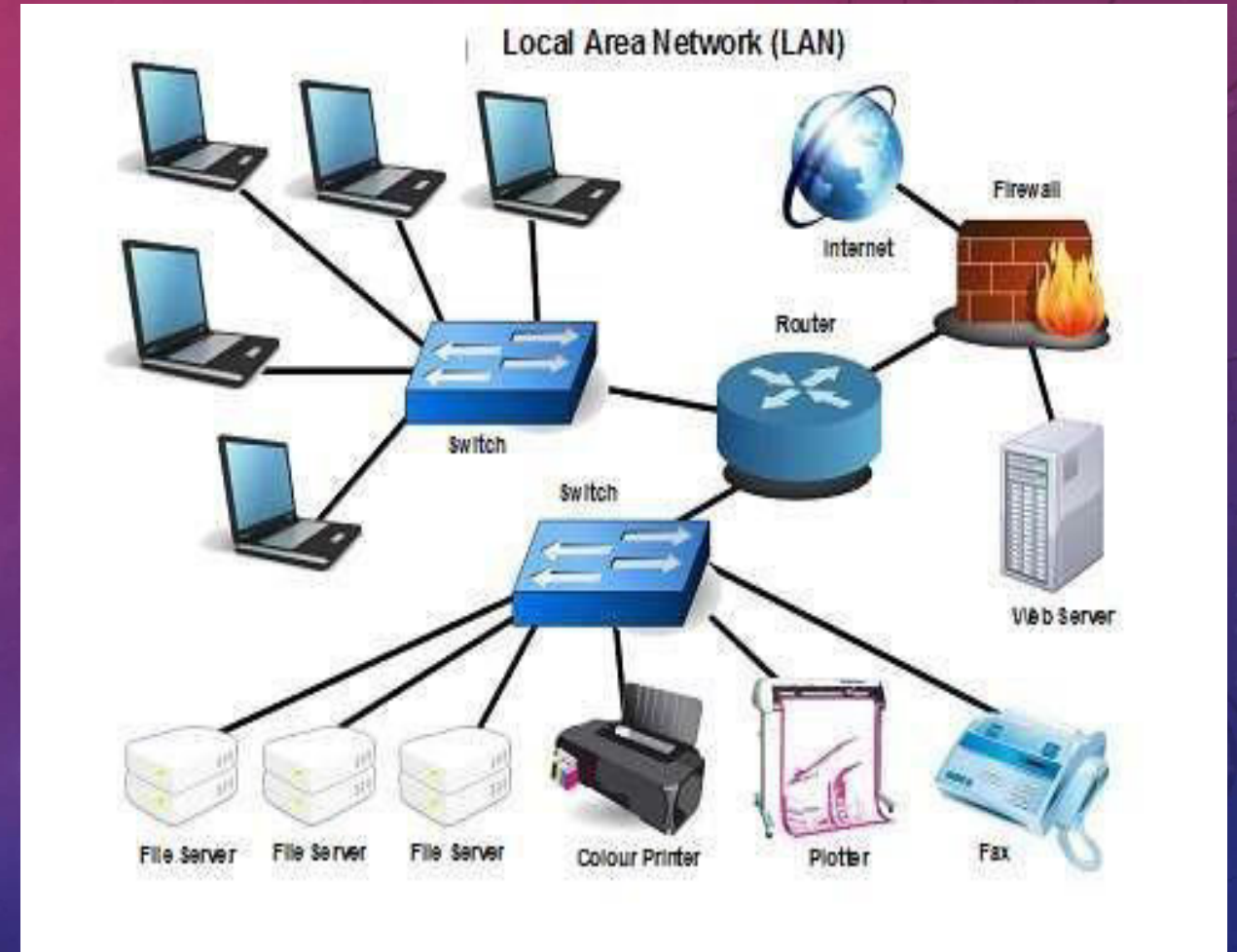
## TYPES OF LOCAL-AREA NETWORKS (LANs)

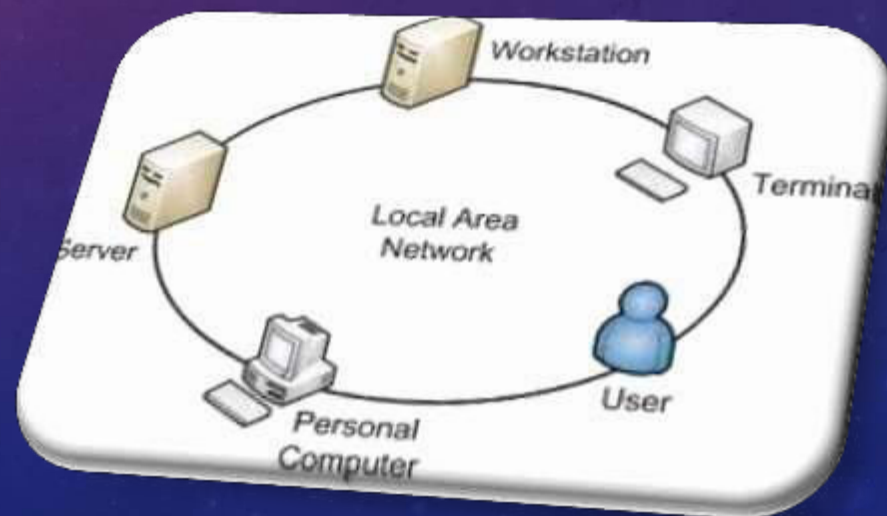
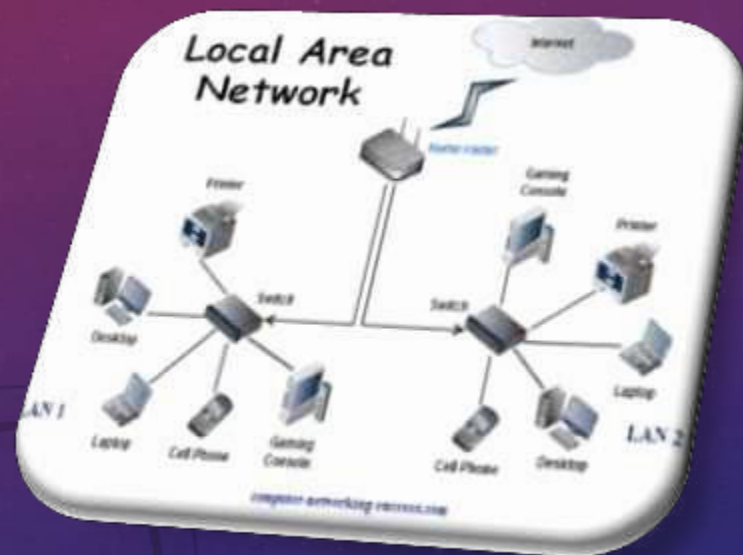
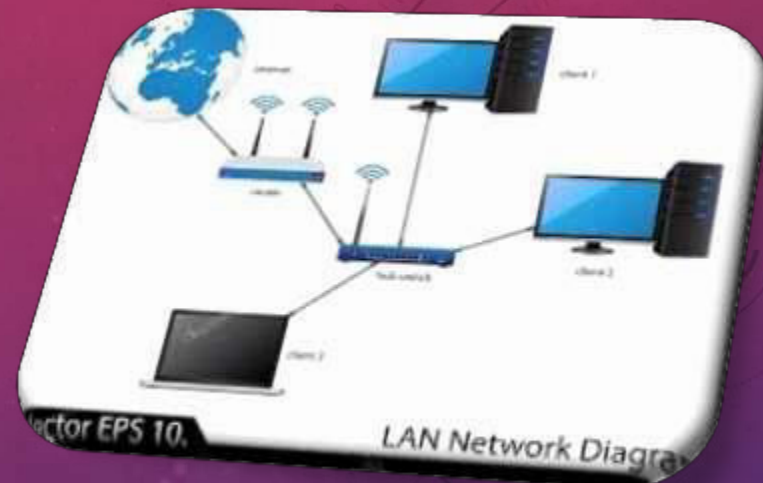
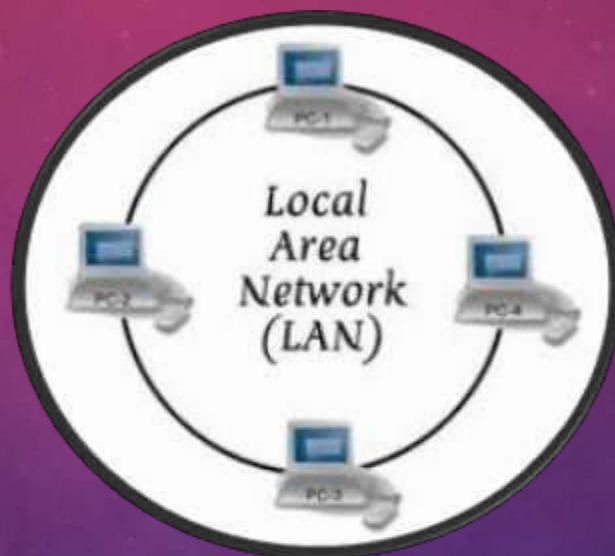
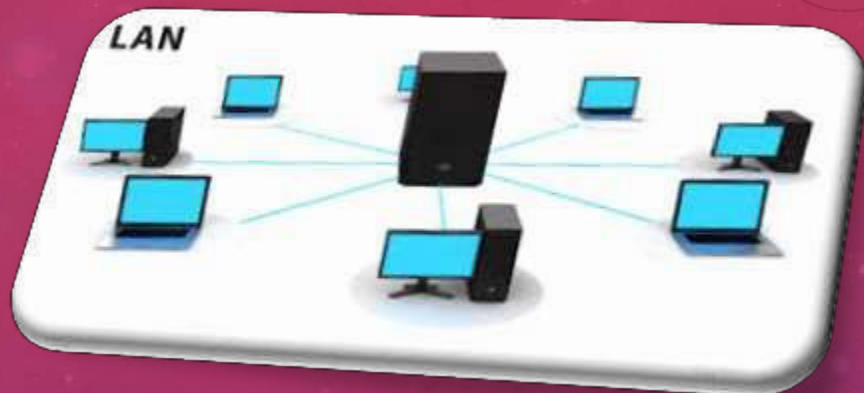
There are many different types of LANs, with Ethernet being the most common for PCs. Most Apple Macintosh networks are based on Apple's AppleTalk network system, which is built into Macintosh computers. The following characteristics differentiate one LAN from another:

**Topology:** The geometric arrangement of devices on the network. For example, devices can be arranged in a ring or in a straight line.

**Protocols:** The rules and encoding specifications for sending data. The protocols also determine whether the network uses a peer-to-peer or client/server architecture.

**Media:** Devices can be connected by twisted-pair wire, coaxial cables, or fiber optic cables. Some networks do without connecting media altogether, communicating instead via radio waves.



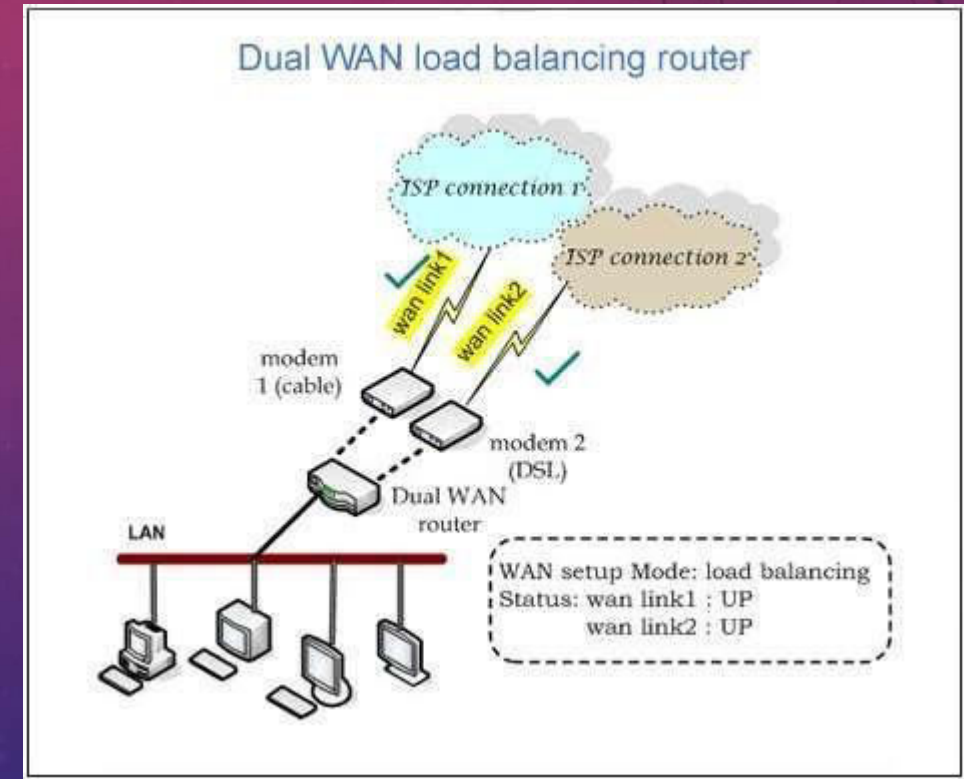


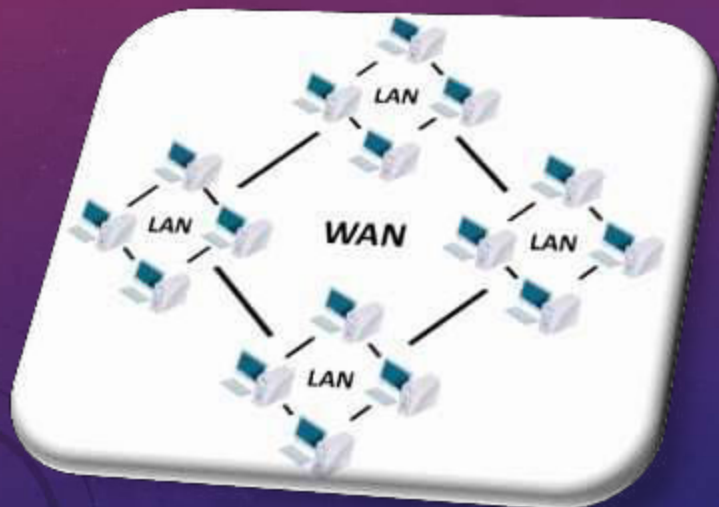
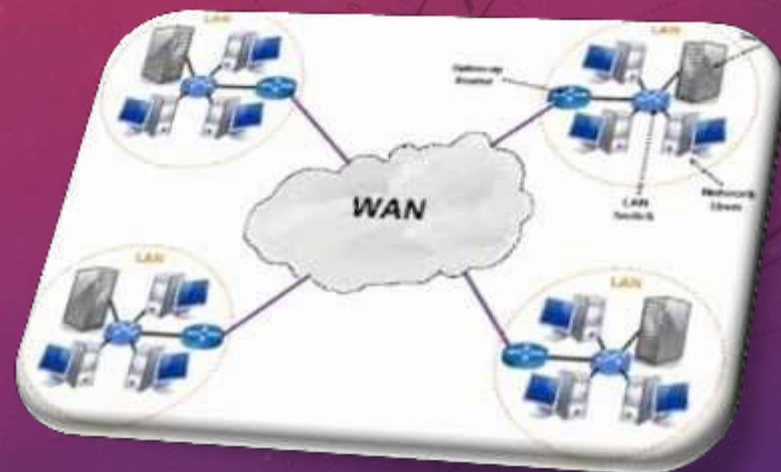
## WIDE AREA NETWORK(WAN)

- A wide area network (WAN) is a network that exists over a large-scale geographical area. A WAN connects different smaller networks, including local area networks (LANs) and metro area networks (MANs). This ensures that computers and users in one location can communicate with computers and users in other locations. WAN implementation can be done either with the help of the public transmission system or a private network.
- A WAN connects more than one LAN and is used for larger geographical areas. WANs are similar to a banking system, where hundreds of branches in different cities are connected with each other in order to share their official data.
- A WAN works in a similar fashion to a LAN, just on a larger scale. Typically, TCP/IP is the protocol used for a WAN in combination with devices such as routers, switches, firewalls, and modems.

## TYPES OF WIDE-AREA NETWORKS

A WAN (Wide Area Network) network can be established in many ways and different types of WAN. WAN can be established using leased lines, circuit switch network, packet switching, frame or packet relay and cell relays. Packet switching is the one of the most common types of establishing a WAN (Wide Area Network) network. In a packet switching method a digital network is established. In this method suitable data blocks are created. These data blocks are known as packets. The data travels in the form of varied bit rate. These bits travel in the form of sequences of packets over the network and their delay is dependent upon the load of traffic over the network. X.25 and frame relays are the most suitable protocols for this kind of network. Leased lines are the other most suitable and quick way to establish a WAN (Wide Area Network).



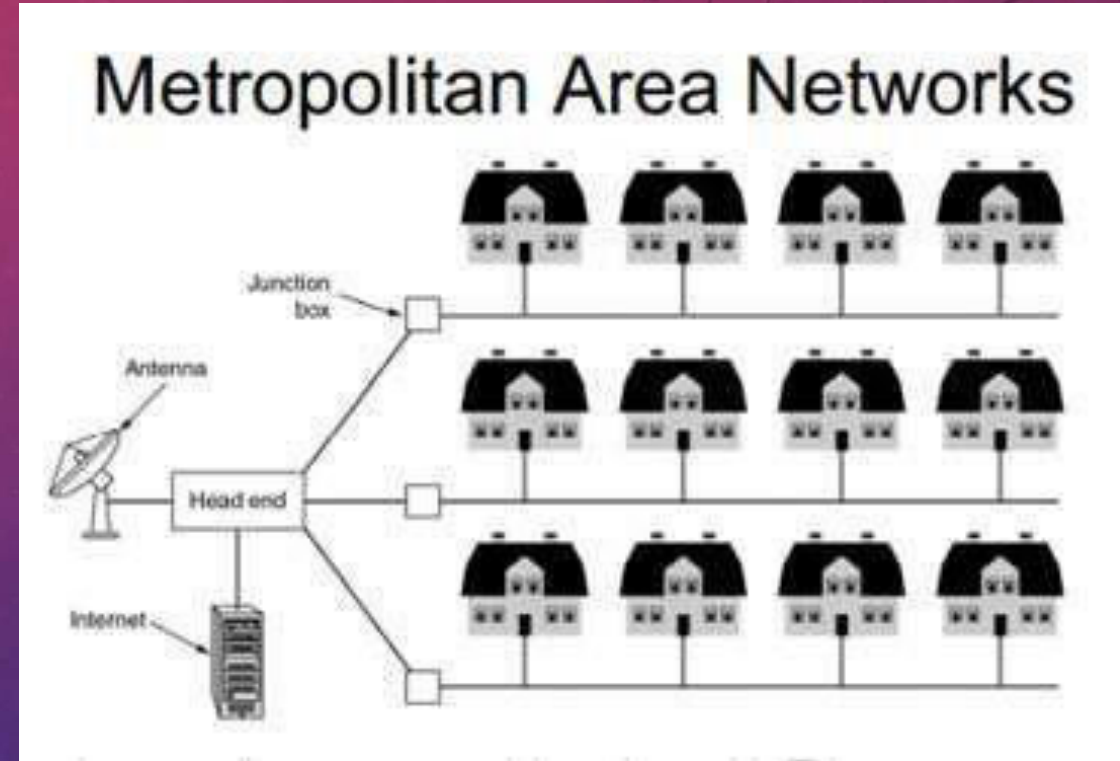


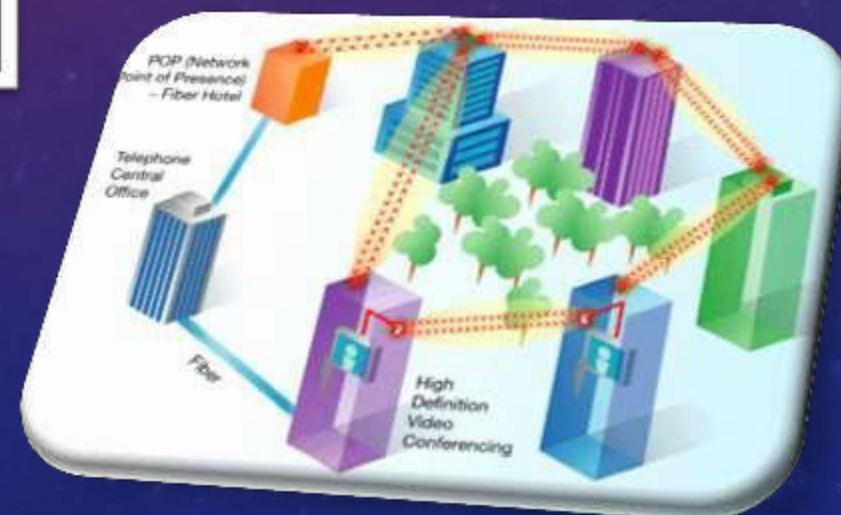
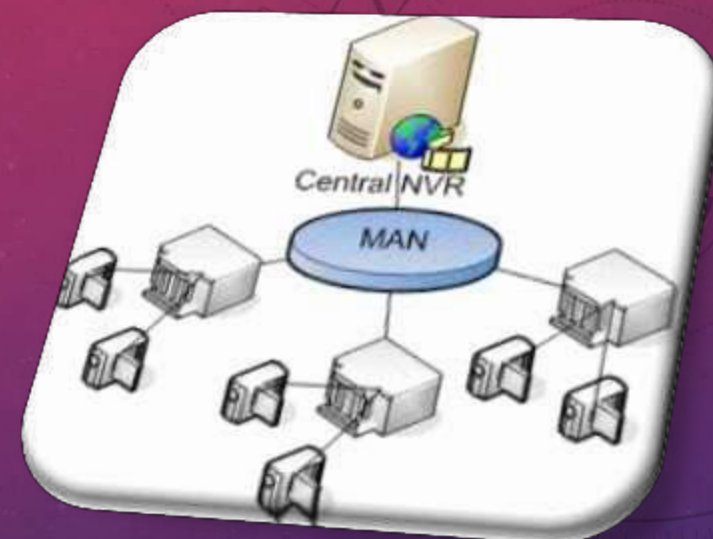
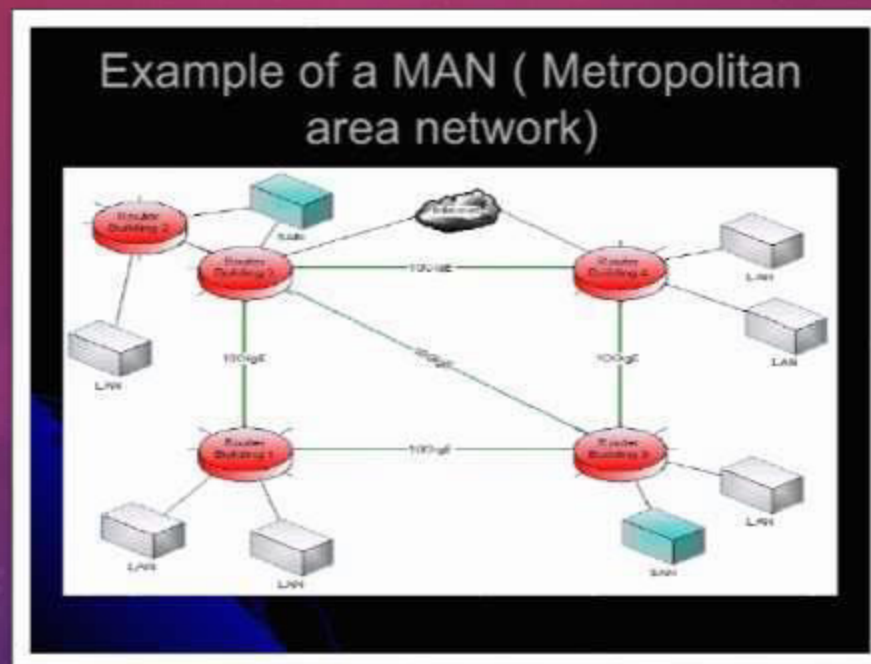
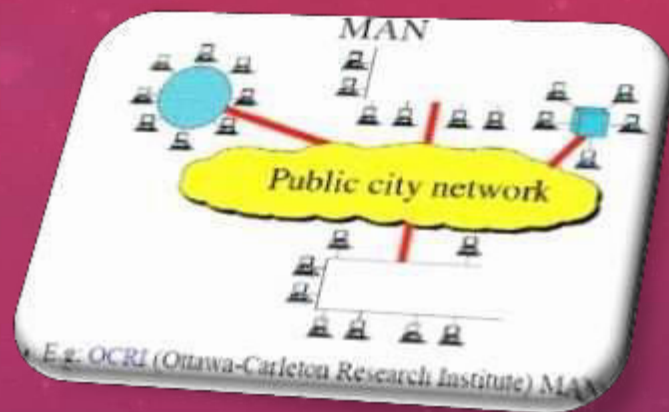
## METROPOLITAN AREA NETWORK (MAN)

- A metropolitan area network (MAN) is a [computer network] that interconnects users with computer resources in a geographic area or region larger than that covered by even a large local area network (**LAN**) but smaller than the area covered by a wide area network (WAN).
- A MAN is ideal for many kinds of network users because it is a medium-size network. MANs are used to build networks with high data connection speeds for cities and towns.
- MANs are extremely efficient and provide fast communication via high-speed carriers, such as fiber optic cables.

## TYPES OF METROPOLITAN -AREA NETWORKS

Most widely used technologies to develop a MAN (Metropolitan Area Network) network are FDDI (fiber distribution data interface), ATM (Asynchronous Transfer Mode) and SMDS (switched multi megabit data service). ATM (Asynchronous Transfer Mode) is the most frequently used of all. ATM (Asynchronous Transfer Mode) is a digital data transfer technology. It was developed in 1980 to improve the transportation of real time data over a single network. ATM (Asynchronous Transfer Mode) works just like cell relay system, where data is separated in the form of fixed equal sized packets and is transferred over time. The purpose of ATM (Asynchronous Transfer Mode) was to access clear audio and video results during a video conferencing. The attributes of ATM have enabled it to become a base of wide area data networking.

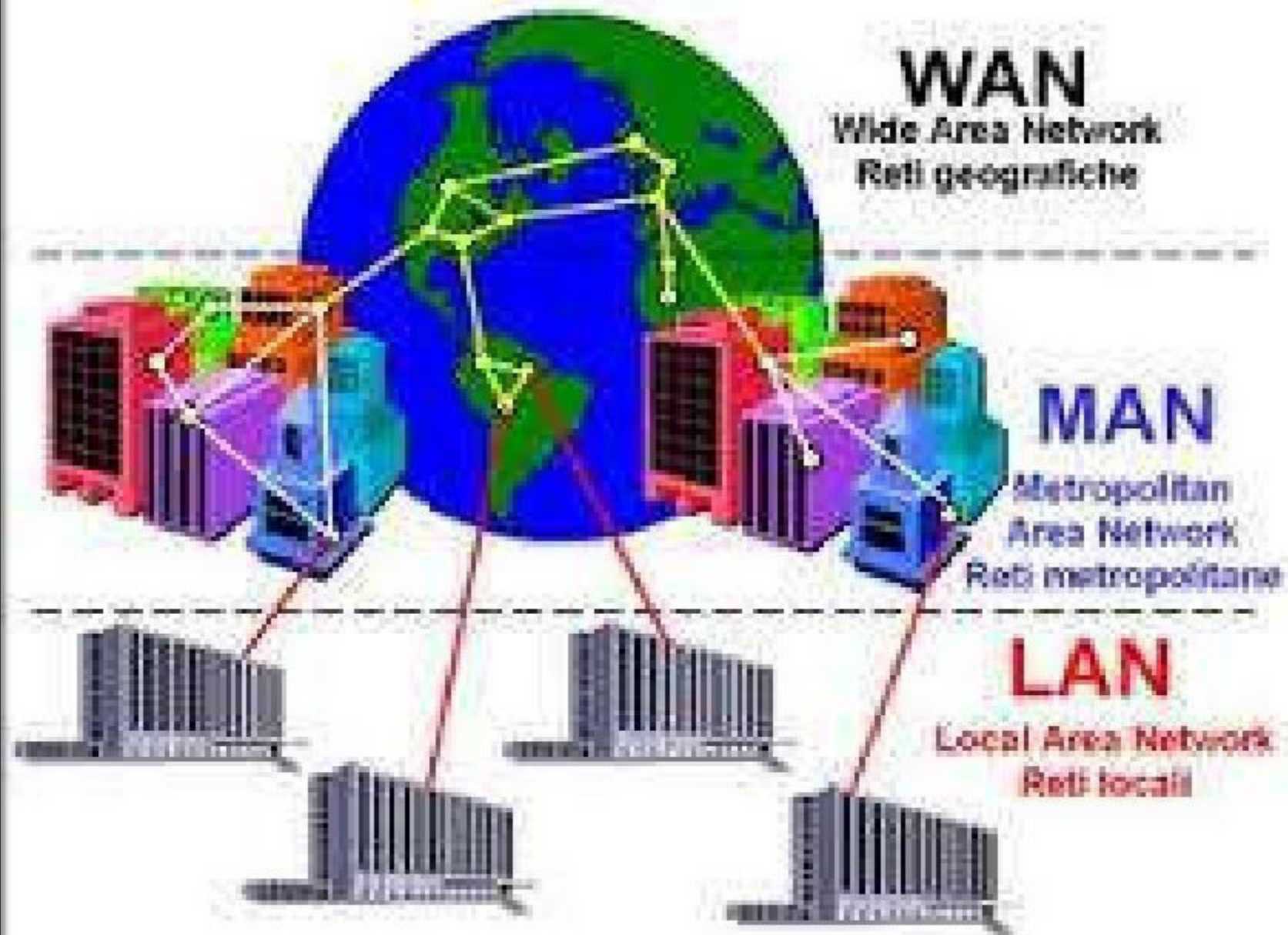




## DIFFERENCE BETWEEN LAN, WAN, MAN

| BASIS OF COMPARISON    | LAN                                                                        | WAN                                                                           | MAN                                                      |
|------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------|
| Expands to             | Local Area Network                                                         | Wide Area Network                                                             | Metropolitan Area Network                                |
| Meaning                | A network that connects a group of computers in a small geographical area. | It spans large locality and connects countries together.<br>Example Internet. | It covers relatively large region such as cities, towns. |
| Ownership of Network   | Private                                                                    | Private or Public                                                             | Private or Public                                        |
| Design and maintenance | Easy                                                                       | Difficult                                                                     | Difficult                                                |

| Speed    | High             | Low                | Moderate          |
|----------|------------------|--------------------|-------------------|
| Used for | College, School, | Country/Continent. | Smal towns, City. |





# THE INTERNET



## History of the internet

- The internet first came about in February 1958 in America
- A man named J.C.R. Licklider saw universal networking as a potential unifying revolution.
- He envisioned a globally interconnected set of computers through which everyone could quickly access data and programs from any site.
- Leonard Kleinrock at MIT published the first paper on packet switching theory in July 1961 and the first book on the subject in 1964.
- The internet began to grow rapidly in the 1970's as the knowledge of networking was growing more and more rapidly.
- It expanded further in the 1980's as the general public were becoming more aware of it.

- The general public gained access to the Internet on July 16, 1986 when the first Freenet came on-line in Cleveland in the U.S.
- Between 1987 and 1988 the number of hosts increased from 10,000 to 60,000
- With the development of the World Wide Web (WWW) in 1991, there was no stopping the rapid expansion of the Internet.
- People only had to download a WWW browser in order to access the info on millions of websites. Countries around the world joined the Internet and registered their two-letter domain names, e.g. ca for Canada.
- The most important development was in 1993 was the release of the first graphic web browser software, Mosaic.

- In 1994 the first junk e-mail or "spam" was sent out this year. The first shopping malls also appeared along with Internet radiostations (mainly rebroadcasting) and the first cyberbank.
- In 1998, the Internet continued to grow, with thousands of new subscribers every day joining the millions already subscribed.



## What is?

- The Internet is a worldwide, publicly accessible network of interconnected computer networks that transmit data using the standard Internet Protocol (IP).
- It's a worldwide network of computers that allows the "sharing" or "networking" of information at remote sites from other academic institutions, research institutes, private companies, government agencies, and individuals.
- It allows people to send and receive data wherever they are in the world if they have internet access.
- Everyone will find that they use the internet in one way or another, whether it is to research some information or simply talk to friends.
- Today, the Web and the Internet allow connectivity from literally everywhere on earth—even ships at sea and in outer space.

## Audience

- The intended audience for the internet is anyone and everyone in the world who has access to a computer; there is no limit to the information that can be found on the internet.
- The internet is used widely by the audience, mostly it is used for researching subjects and talking online. However it has many other uses, one advantage of the internet is that it can be used for advertising products and companies. If you put an advert for a product on a site that is visited regularly it will become more popular.
- The audience uses the internet for personal use to make their life easier and faster. The internet can enable the audience to find information within minutes and also allows them to talk to people all over the world wherever they are.

How does it affect the audience?

- It affects the audience in a good way because it makes their lives easier and less problematic
- It enables them to do well in school and in work as they can access information and advice easily
- However it can also change a person in the way they think and their lives in general, as the internet allows them to access anything from anywhere in the world and this isn't always a good thing as extreme violence etc is promoted.
- The audience use the internet for many different ways, in which are explained on the next page. The internet is used in places such as school, at home and at work.

## Why is it used?

The internet is used for many different things such as:

- **Talking to friends**- using programmes such as msn and myspace (web2)
- **Online shopping**- buying items from the internet without leaving your home using sites such as [www.tesco.com](http://www.tesco.com)
- **Watching Videos**- using websites such as [www.youtube.com](http://www.youtube.com)
- **Research**- using search engines such as [www.google.com](http://www.google.com)
- **Downloading/listening to music**- using programmes such as lime wire or sites such as [www.myspace.com](http://www.myspace.com)

## Advantages and Disadvantages

The internet has many more advantages than it does disadvantages according to research.

Advantages:

- ✓ Allow easy and quick access to information
- ✓ Enables user to access files and information from any computer anywhere in the world
- ✓ People can check financial information, purchase products, talk to people and much more online
- ✓ Downloading music - quick and easy
- ✓ User can send information quickly from computer to computer

### Disadvantages:

- ✓ It promotes violence and bad behaviour within society, as it contains violent videos and other disturbing images
- ✓ It can encourage negative activities such as pornography and paedophilia, these things can now be explored more openly due to the internet
- ✓ It stops children from going out, as they wish to stay home and use the internet to talk to friends etc

### The value of the internet for media institutions

The internet is a great advantage to different institutions around the world.

It enable them to:

- Advertise online which gives them further promotion
- Allow customers etc to make online accounts and buy products online
- They can use the internet to create their own website which can give clients the latest news etc
- Allow them to send email etc to institutions in other countries to also give further promotion and development
- They can download information from the internet on to their computer systems quick and easily

Questions:-

Q1. Differentiate between internet and intranet.

Q2. Describe DNS.

Q3. Explain the structure of URL.

Q4. Explain the purpose of web Browser Software.



I think the internet will develop in the future in the following ways:

- It will become cheaper for people to use
- It will become much faster and information will be able to be accessed more quickly
- More and more information will be added to it, which will make it the world's biggest technology resource
- It will continue to grow and as it becomes cheaper people in less economically developed countries will gain access to it, further developing their education and access to information
- Overall, the internet will become even more advanced in technology, making it cheaper, faster and more widely accessed.

# ALGORITHMS AND FLOWCHARTS

# ALGORITHMS AND FLOWCHARTS

- A typical programming task can be divided into two phases:
- ***Problem solving phase***
  - produce an ordered sequence of steps that describe solution of problem
  - this sequence of steps is called an ***algorithm***
- ***Implementation phase***
  - implement the program in some programming language

# Steps in Problem Solving

- First produce a general algorithm (one can use ***pseudocode***)
- Refine the algorithm successively to get step by step detailed ***algorithm*** that is very close to a computer language.
- ***Pseudocode*** is an artificial and informal language that helps programmers develop algorithms. Pseudocode is very similar to everyday English.

# Pseudocode & Algorithm

- **Example 1:** Write an algorithm to determine a student's final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks.

# Pseudocode & Algorithm

## **Pseudocode:**

- *Input a set of 4 marks*
- *Calculate their average by summing and dividing by 4*
- *if average is below 50*  
    *Print "FAIL"*  
    *else*  
        *Print "PASS"*

# Pseudocode & Algorithm

- Detailed Algorithm
- Step 1: Input M1,M2,M3,M4
- Step 2:  $\text{GRADE} \leftarrow (M1+M2+M3+M4)/4$
- Step 3: if (GRADE < 50) then  
                    Print "FAIL"  
                    else  
                        Print "PASS"  
                    endif

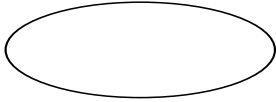
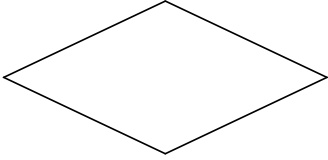
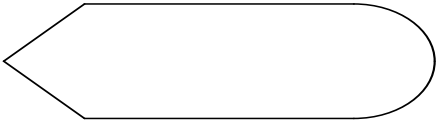
# The Flowchart

## A Flowchart

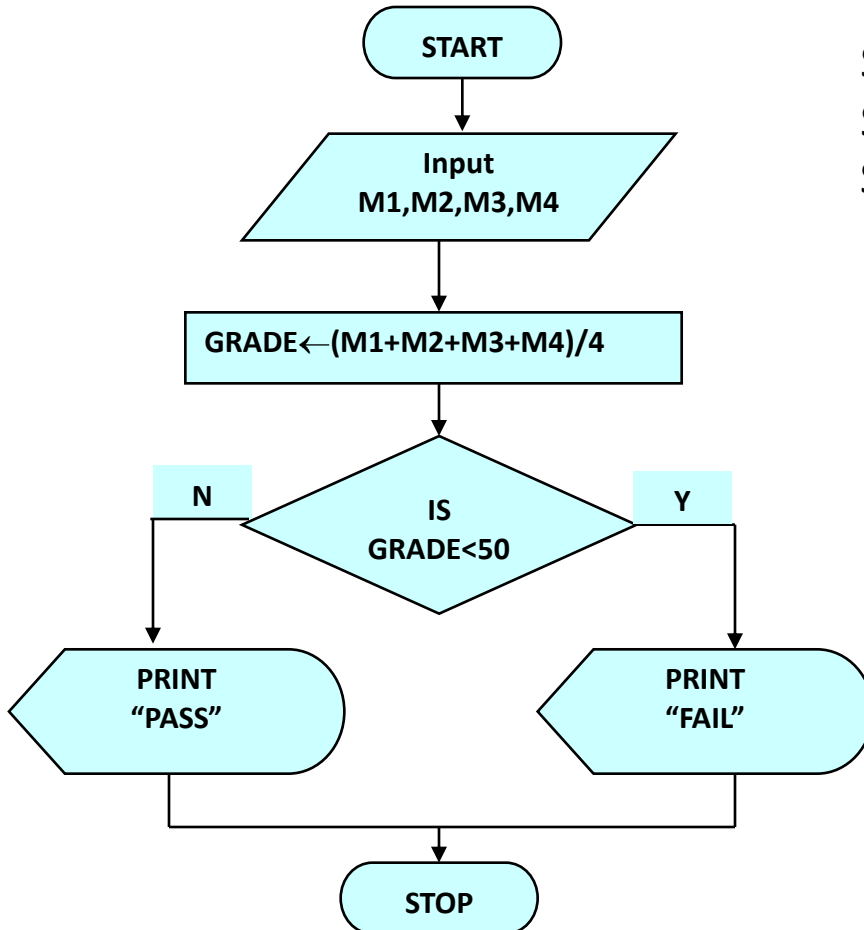
- shows logic of an algorithm
- emphasizes individual steps and their interconnections
- e.g. control flow from one action to the next

# Flowchart Symbols

## Basic

| Name          | Symbol                                                                              | Use in Flowchart                                                                                                       |
|---------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Oval          |    | Denotes the beginning or end of the program                                                                            |
| Parallelogram |    | Denotes an input operation                                                                                             |
| Rectangle     |    | Denotes a process to be carried out<br>e.g. addition, subtraction, division etc.                                       |
| Diamond       |   | Denotes a decision (or branch) to be made.<br>The program should continue along one of two routes. (e.g. IF/THEN/ELSE) |
| Hybrid        |  | Denotes an output operation                                                                                            |
| Flow line     |  | Denotes the direction of logic flow in the program                                                                     |

# Example



Step 1: Input M1,M2,M3,M4

Step 2:  $GRADE \leftarrow (M1 + M2 + M3 + M4) / 4$

Step 3: if (GRADE < 50) then  
Print "FAIL"

else

Print "PASS"

endif

## Example 2

- Write an algorithm and draw a flowchart to convert the length in feet to centimeter.

### **Pseudocode:**

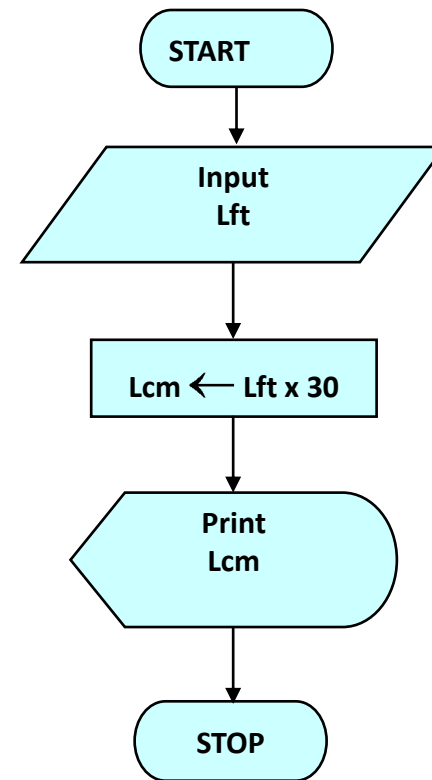
- *Input the length in feet (Lft)*
- *Calculate the length in cm (Lcm) by multiplying LFT with 30*
- *Print length in cm (LCM)*

# Example 2

## Algorithm

- Step 1: Input Lft
- Step 2:  $Lcm \leftarrow Lft \times 30$
- Step 3: Print Lcm

## Flowchart



# Example 3

**Write an algorithm and draw a flowchart that will read the two sides of a rectangle and calculate its area.**

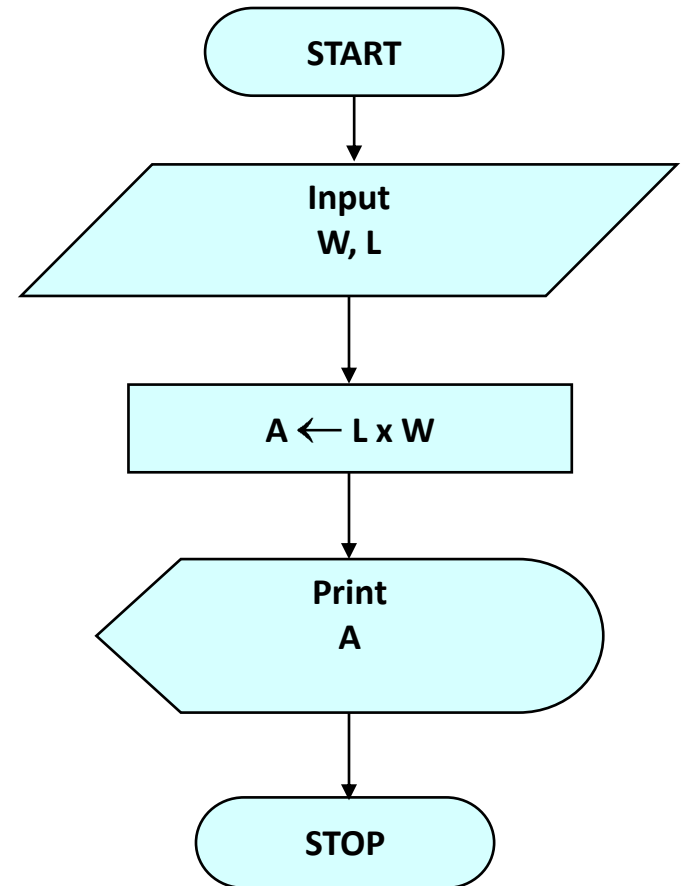
## **Pseudocode**

- *Input the width ( $W$ ) and Length ( $L$ ) of a rectangle*
- *Calculate the area ( $A$ ) by multiplying  $L$  with  $W$*
- *Print  $A$*

# Example 3

## Algorithm

- Step 1: Input W,L
- Step 2:  $A \leftarrow L \times W$
- Step 3: Print A



# Example 4

- Write an algorithm and draw a flowchart that will calculate the roots of a quadratic equation

$$ax^2 + bx + c = 0$$

- $X = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$
- $X = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

# Example 4

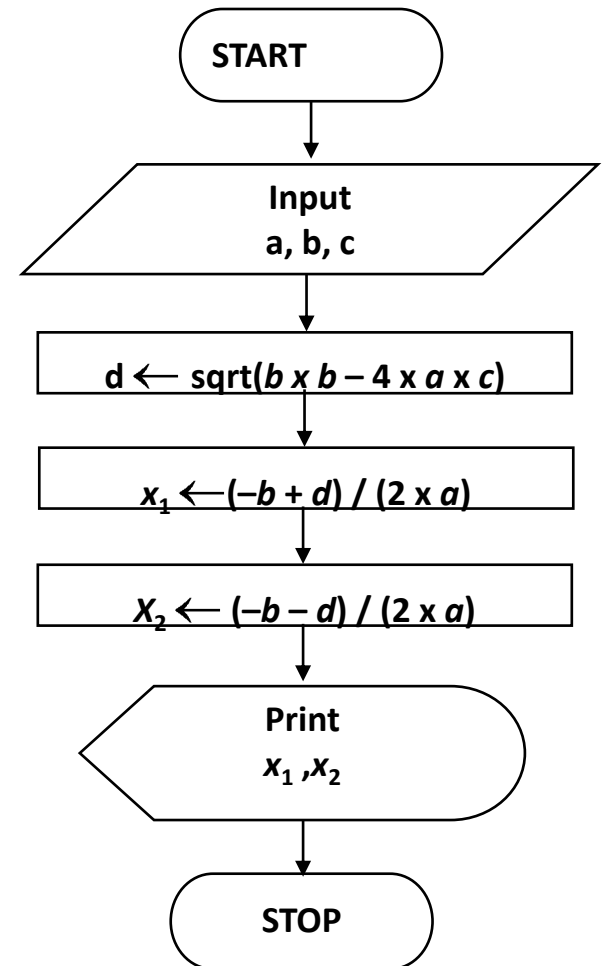
## Pseudocode:

- *Input the coefficients ( $a$ ,  $b$ ,  $c$ ) of the quadratic equation*
- *Calculate **d***
- *Calculate **x1***
- *Calculate **x2***
- *Print  $x1$  and  $x2$*

# Example 4

- **Algorithm:**

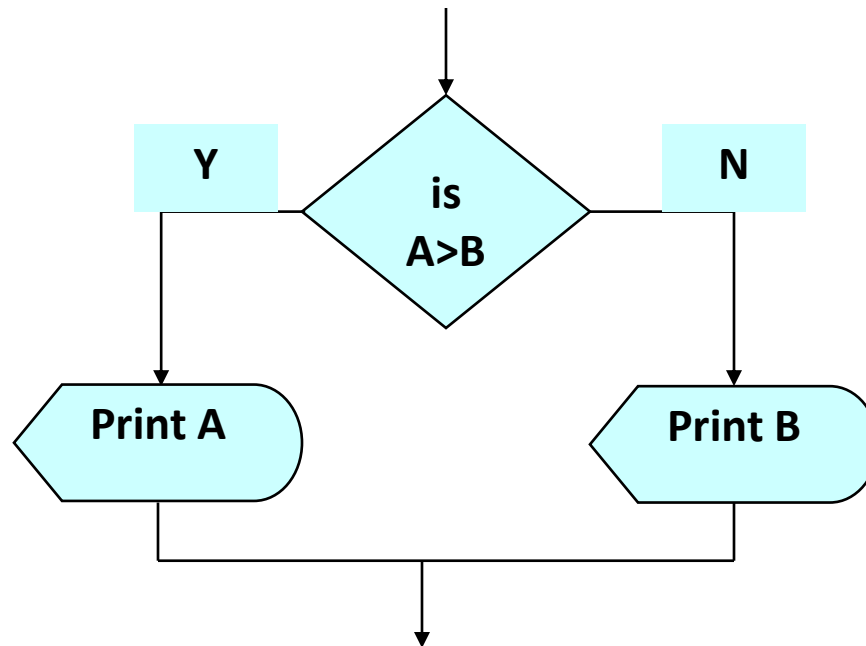
- Step 1: Input a, b, c
- Step 2:  $d \leftarrow \text{sqrt}(b \times b - 4 \times a \times c)$
- Step 3:  $x_1 \leftarrow (-b + d) / (2 \times a)$
- Step 4:  $x_2 \leftarrow (-b - d) / (2 \times a)$
- Step 5: Print x1, x2



# DECISION STRUCTURES

- The expression  $A > B$  is a logical expression
- *it describes a **condition** we want to test*
- ***if  $A > B$  is true (if  $A$  is greater than  $B$ )** we take the action on left*
- print the value of A
- ***if  $A > B$  is false (if  $A$  is not greater than  $B$ )** we take the action on right*
- print the value of B

# DECISION STRUCTURES



# IF–THEN–ELSE STRUCTURE

- The structure is as follows

*If condition then*

*true alternative*

*else*

*false alternative*

*endif*

# IF-THEN-ELSE STRUCTURE

- The algorithm for the flowchart is as follows:

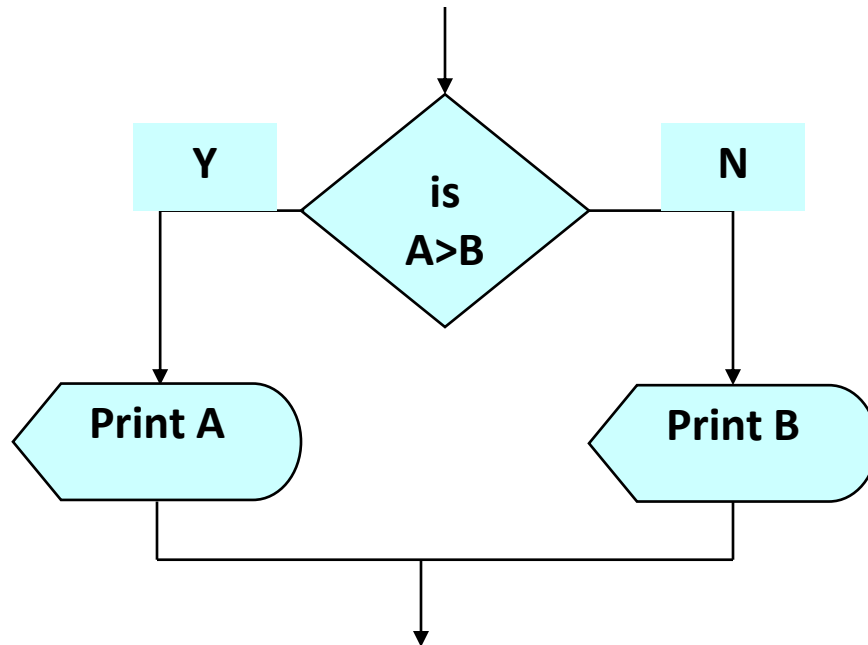
***If  $A > B$  then***

***print A***

***else***

***print B***

***endif***



# Relational Operators

| Relational Operators |                          |
|----------------------|--------------------------|
| Operator             | Description              |
| $>$                  | Greater than             |
| $<$                  | Less than                |
| $=$                  | Equal to                 |
| $\geq$               | Greater than or equal to |
| $\leq$               | Less than or equal to    |
| $\neq$               | Not equal to             |

# Example 5

- Write an algorithm that reads two values, determines the largest value and prints the largest value with an identifying message.

## **ALGORITHM**

Step 1:            *Input* VALUE1, VALUE2

Step 2:            *if* (VALUE1 > VALUE2) *then*

                    MAX ← VALUE1

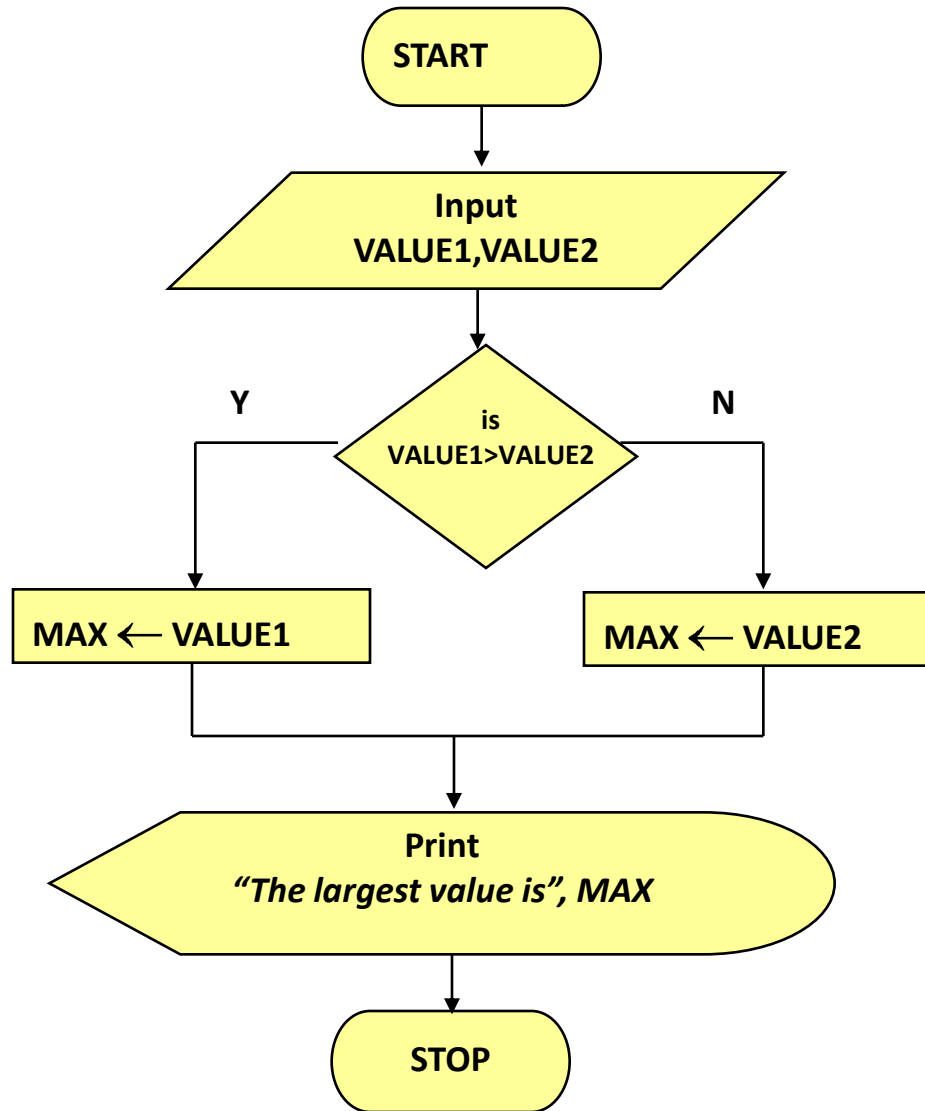
*else*

                    MAX ← VALUE2

*endif*

Step 3:            *Print* “The largest value is”, MAX

# Example 5



# NESTED IFS

- One of the alternatives within an IF–THEN–ELSE statement
  - may involve further IF–THEN–ELSE statement

## Example 6

- Write an algorithm that reads **three** numbers and prints the value of the largest number.

# Example 6

Step 1: *Input*       $N1, N2, N3$

Step 2: *if* ( $N1 > N2$ ) *then*

*if* ( $N1 > N3$ ) *then*

$MAX \leftarrow N1$       [ $N1 > N2, N1 > N3$ ]

*else*

$MAX \leftarrow N3$       [ $N3 > N1 > N2$ ]

*endif*

*else*

*if* ( $N2 > N3$ ) *then*

$MAX \leftarrow N2$       [ $N2 > N1, N2 > N3$ ]

*else*

$MAX \leftarrow N3$       [ $N3 > N2 > N1$ ]

*endif*

*endif*

Step 3: *Print* “The largest number is”,  $MAX$

# Example 6

- **Flowchart: Draw the flowchart of the above Algorithm.**

# Example 7

- Write an algorithm and draw a flowchart to
  - a) read an employee name (NAME), overtime hours worked (OVERTIME), hours absent (ABSENT) and
  - b) determine the bonus payment (PAYMENT).

# Example 7

| Bonus Schedule                   |            |
|----------------------------------|------------|
| OVERTIME – $(2/3)*\text{ABSENT}$ | Bonus Paid |
| >40 hours                        | \$50       |
| >30 but $\leq$ 40 hours          | \$40       |
| >20 but $\leq$ 30 hours          | \$30       |
| >10 but $\leq$ 20 hours          | \$20       |
| $\leq$ 10 hours                  | \$10       |

Step 1: *Input* NAME,OVERTIME,ABSENT

Step 2: *if* (OVERTIME–(2/3)\*ABSENT > 40) *then*

PAYMENT  $\leftarrow$  50

*else if* (OVERTIME–(2/3)\*ABSENT > 30) *then*

PAYMENT  $\leftarrow$  40

*else if* (OVERTIME–(2/3)\*ABSENT > 20) *then*

PAYMENT  $\leftarrow$  30

*else if* (OVERTIME–(2/3)\*ABSENT > 10) *then*

PAYMENT  $\leftarrow$  20

*else*

PAYMENT  $\leftarrow$  10

*endif*

Step 3: *Print* “Bonus for”, NAME “is \$”, PAYMENT

# Example 7

- **Flowchart: Draw the flowchart of the above algorithm?**

# Objectives

- In this chapter you will be able to:
- Introduce common words, keywords, and meaningful names when writing pseudocode
- Define the three basic control structures as set out in the Structure Theorem
- Illustrate the three basic control structures using pseudocode

# How to Write Pseudocode

- When designing a solution algorithm, you need to keep in mind that a computer will eventually perform the set of instructions written
- If you use words and phrases in the pseudocode which are in line with basic computer operations, the translation from pseudocode algorithm to a specific programming language becomes quite simple

# Six Basic Computer Operations

## 1 A computer can receive information

- When a computer is required to receive information or input from a particular source, whether it is a terminal, a disk or any other device, the verbs **Read** and **Get** are used in pseudocode

**Read** => Input from a record

**Get** => Input from keyboard

### Example pseudocode

```
Read student name
Get system data
Read number1, number2
Get tax_code
```

# Six Basic Computer Operations

## 1 A computer can receive information

- Usually an output **Prompt** instruction is required before an input **Get** instruction

### **Example pseudocode**

```
Prompt for student_mark  
Get student_mark
```

# Six Basic Computer Operations

## 2 A computer can put out information

- When a computer is required to supply information or output to a device, the verbs **Print**, **Write**, **Put**, **Output**, or **Display** are used in pseudocode
- **Print** => send output to printer
- **Write** => send out to file
- **Put**, **Output**, **Display** => send to screen

### Example pseudocode

Print 'Program Completed'  
Write customer record to master file  
Output total tax  
Display 'End of data'

# Six Basic Computer Operations

## 3 A computer can perform arithmetic

- Most programs require the computer to perform some sort of mathematical calculation, or formula, and for these, a programmer may use either actual mathematical symbols or the words for those symbols
- To be consistent with high-level programming languages, the following symbols can be written in pseudocode:
  - + for Add                      - for Subtract
  - \* for Multiply                      / for Divide                      ( ) for Parentheses
- When writing mathematical calculations for the computer, standard mathematical 'order of operations' applies to pseudocode and most computer languages

# Six Basic Computer Operations

## 4 A computer can assign a value to a variable or memory location

- There are three cases where you may write pseudocode to assign a value to a variable or memory location:
  1. To give data an initial value in pseudocode, the verbs **Initialize** or **Set** are used
  2. To assign a value as a result of some processing the symbols '=' or '←' are written
  3. To keep a variable for later use, the verbs **Save** or **Store** are used

# Six Basic Computer Operations

## 4 A computer can assign a value to a variable or memory location

### Example pseudocode

Initialize total\_price to zero

Set student\_count to zero

Total\_price = cost\_price + sales\_tax

Total\_price  $\leftarrow$  cost\_price + sales\_tax

Store customer\_num in last\_customer\_num

# Six Basic Computer Operations

## 5 A computer can compare two variables and select one or two alternate actions

- An important computer operation available to the programmer is the ability to compare two variables and then, as a result of the comparison, select one of two alternate actions
- To represent this operation in pseudocode, special keywords are used: **IF**, **THEN**, and **ELSE**

# Six Basic Computer Operations

## 6 A computer can repeat a group of actions

- When there is a sequence of processing steps that need to be repeated, two special keywords, **DOWHILE** and **ENDDO**, are used in pseudocode
- The condition for the repetition of a group of actions is established in the **DOWHILE** clause, and the actions to be repeated are listed beneath it

# Meaningful Names

- All names should be meaningful
- A name given to a variable is simply a method of identifying a particular storage location in the computer
- The uniqueness of a name will differentiate it from other locations
- Often a name describes the type of data stored in a particular variable
- Most programming languages do not tolerate a space in a variable name, as a space would signal the end of the variable name and thus imply that there were two variables

# The Structure Theorem

- The Structure Theorem states that it is possible to write any computer program by using only three basic control structures that are easily represented in pseudocode:
  - Sequence
  - Selection
  - Repetition

# The Three Basic Control Structures

## 1 Sequence

- The sequence control structure is the straightforward execution of one processing step after another
- In pseudocode, we represent this construct as a sequence of pseudocode statements

## 2 Selection

- The selection control structure is the presentation of a condition and the choice between two actions; the choice depends on whether the condition is true or false
- In pseudocode, selection is represented by the keywords **IF**, **THEN**, **ELSE**, and **ENDIF**

# The Three Basic Control Structures

## 3 Repetition

- The repetition control structure can be defined as the presentation of a set of instructions to be performed repeatedly, as long as a condition is true
- The basic idea of repetitive code is that a block of statements is executed again and again, until a terminating condition occurs
- This construct represents the sixth basic computer operation, namely to repeat a group of actions

# Summary

- In this chapter, six basic computer operations were listed, along with pseudocode words and keywords to represent them
- These operations were: to receive information, put out information, perform arithmetic, assign a value to a variable, decide between two alternate actions, and repeat a group of actions
- The Structure Theorem was introduced; it states that it is possible to write any computer program by using only three basic control structures: sequence, selection, and repetition

## Someone Stole a Cookie from the Cookie Jar

Problem: Momma had just filled the cookie jar when the 3 children went to bed. That night one child woke up, ate half of the cookies and went back to bed. Later, the second child woke up, ate half of the remaining cookies, and went back to bed. Still later, the third child woke up, ate half of the remaining cookies, leaving 3 cookies in the jar. How many cookies were in the jar to begin with?

## Someone Stole a Cookie from the Cookie Jar (cont'd)

- Information available:
  - Three children
  - Each one ate half of the cookies
  - Three cookies remaining
- Information needed:
  - Original number of cookies
- Calculations:
  - For each child, multiply the number of remaining cookies by two.

# Specific Solution to the Problem

- First, we solve the specific problem to help us identify the steps.
  - 3 cookies left  $\times 2 = 6$  cookies left after 2nd child
  - $6 \times 2 = 12$  cookies left after 1st child
  - $12 \times 2 = 24 =$  original number of cookies

# A Generic Algorithm

- What is a **generic algorithm** for this problem?

An algorithm that will work with any number of remaining cookies

AND

that will work with any number of children.

# Generic Algorithm for Cookie Problem

- Get number of children.
- Get number of cookies remaining.
- While there are still children that have not raided the cookie jar, multiply the number of cookies by 2 and reduce the number of children by 1.
- Display the original number of cookies.

# Test The Generic Algorithm

- Try the algorithm on paper with:
  - Four children and six cookies remaining.
  - Two children with two cookies remaining.
- If you did not get the correct answer, modify the algorithm so that you get the correct answer.

# Pseudocode

- When we broke down the previous problem into steps, we expressed each step as an English phrase.
- We can think of this as writing **pseudocode** for the problem.
- Typically, pseudocode is a combination of English phrases and formulas.

# Pseudocode (con't)

- Pseudocode is used in
  - designing algorithms
  - communicating an algorithm to the customer
  - converting an algorithm to code (used by the programmer)
  - **debugging** logic (semantic) errors in a solution before coding (**hand tracing**)
- Let's write the Cookie Problem algorithm using a more formal pseudocode and being more precise.

# Improved Pseudocode

Display "Enter the number of children: "

Read <number of children>

Display "Enter the number of cookies remaining: "

Read <cookies remaining>

<original cookies> = <cookies remaining>

While (<number of children> > 0)

    <original cookies> = <original cookies> X 2

    <number of children> = <number of children> - 1

End\_While

Display "Original number of cookies = ", <original cookies>